Hasan Hüseyin Yurdagül

SENIOR MACHINE LEARNING ENGINEER

Adana, Türkiye
Open to relocation and remote work
hhyurdagul@gmail.com
hhyurdagul.com
github.com/hhyurdagul
linkedin.com/in/hhyurdagul

PROFILE

Senior Machine Learning Engineer and hands-on Technical Lead with 5+ years delivering forecasting, computer-vision, NLP/LLM, and recommender systems from evaluation design through production. Leads AI R&D for products serving municipalities, ministries, and utilities, combining model development, retrieval and LLM tooling with APIs, CI/CD, and cloud deployment.

EXPERIENCE

Universal Software

Istanbul-based · Remote

Technical Lead, AI R&D

Jan 2024 - Present

- Lead the hands-on AI R&D roadmap, architecture reviews, delivery planning, and stakeholder coordination; mentor up to 5 engineers building ML products for municipalities, ministries, and utilities.
- Architected a human-in-the-loop support intelligence platform for 60+ municipalities over 132K support events / 29K tickets, delivering evidence retrieval (97% Recall@5 offline), answer drafts, and automated routing; cut median first-response time by 50% (MdsAI).
- Co-lead a TÜBİTAK 1832 national R&D grant with hydraulic engineers; designed attention-based seq2seq river-discharge forecasting with event-based backtests and FastAPI serving for 8 gauging stations (Teus).

AI/ML Engineer → Senior AI/ML Engineer

Apr 2021 - Dec 2023 · promoted Jan 2023

- Shipped a real-time Enerjisa field-photo validation pipeline combining YOLOv8 detection with blur, duplication, and empty-frame checks, reducing invalid submissions reaching inventory review by 30%.
- Designed IoT-driven waste-collection routing that converted live bin-fullness and fire/flood sensor events into priority routes, reducing planned distance by 18% in historical simulations.
- Built predictive-maintenance classifiers for Enerjisa assets and CNC assembly lines, identifying 82% of failures within a 7-day warning window.
- Developed LSTM pipelines for municipal water-demand and industrial gas/electricity forecasting, improving MAPE by 21% relative to seasonal-naive baselines.

OxyAI

Remote, Türkiye

AI/ML Engineer (Part-time Contract)

May 2021 - Jan 2023

- Built production recommenders for the online art gallery **Collectors** and audiobook app **Dinlebi**, combining multimodal and collaborative-filtering approaches with cold-start onboarding; increased recommendation CTR by 14%, and both systems remained in production for 3+ years.

CONSULTANCY

Independent ML Consultant

Sep 2020 - Present

Part-time, alongside employment · Remote, Türkiye

Selected consulting work

- **Moka** (fintech, ongoing): built churn/lifetime-value scoring, price-quote prediction, and financial forecasting behind FastAPI with validated feature contracts, on Azure, Docker, and Kubernetes; the merchant-churn model reached 0.93 F1 / 0.95 ROC-AUC in nested cross-validation.
- **Ikas**: built call-center planning APIs that turn call-volume and handling-time forecasts into Erlang-based staffing requirements, alongside demand and revenue forecasting services.
- **Inveon**: built semantic product search and a mixture-of-experts classifier reaching 91% top-1 accuracy across 50+ subcategories, in production since 2023, plus order cancellation/return prediction APIs.
- **MLStudio**: built a reusable desktop forecasting workbench used across 30+ organizations, cutting time to a first validated model from 10 working days to 2; later rewrote it in Streamlit/PyTorch with time-aware validation and versioned model bundles.
- **Smartiks** (2023 - 2026): led a three-person AI/R&D team on predictive modeling, segmentation, and drone-based bird detection, and wrote the technical parts of ITEA consortium proposals.

SELECTED PROJECTS

Teus - River-Discharge Forecasting hhyurdagul.com/projects/teus

> Built a river-discharge forecasting pipeline for 8 gauging stations: expanding-window backtests on held-out flood events, attention-based seq2seq models compared against persistence and foundation-model baselines, reproducible runs with champion promotion, and FastAPI serving that validates input contracts before hot-swapping checkpoints.
> Python · PyTorch · scikit-learn · Polars · FastAPI · TimesFM

Fishometry - Single-Image Fish-Length Estimation (MSc thesis) hhyurdagul.com/projects/fishometry

> Estimates fish length from one photo without depth sensors or reference objects: YOLO landmarks, Depth Anything V2, and SAM segmentation feed DINOv2 features into a species-aware ridge regressor. Reached 5.53 cm MAE vs. 8.76 cm for a species-mean baseline (37% lower) on 348 held-out outdoor photos.
> Python · PyTorch · DINOv2 · YOLO11 · SAM · Depth Anything V2 · scikit-learn · Polars

Muninn - LLM Agent for a Markdown Knowledge Vault github.com/hhyurdagul/muninn

> Built a Rust agent that reads, searches, and edits a personal notes vault through a streaming tool-call loop over OpenAI-compatible APIs. Exposes CLI, TUI, and an MCP server for Claude Code/Codex, sandboxes file access to the vault, shows edits for review, and records every change in a journal with conflict-aware undo.
> Rust · Tokio · Ratatui · MCP (JSON-RPC) · OpenAI-compatible APIs

SKILLS

- **Programming & Data:** Python, Rust, SQL, Bash, Polars
- **ML Frameworks:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost/LightGBM, sentence-transformers, BERTopic
- **ML Systems & Deployment:** FastAPI, Docker, Kubernetes, Azure, CI/CD, model versioning, MCP
- **Specializations:** Time-series forecasting (LSTM, seq2seq + attention, Transformers), NLP/LLMs (embeddings, RAG, semantic search, tool-calling agents), computer vision (YOLO, segmentation, depth estimation, DINOv2), churn/CLTV, recommenders, anomaly detection
- **Languages:** Turkish (native), English (C1; IELTS 7.5)

EDUCATION

Çukurova University Adana, Türkiye

MSc in Computer Engineering Aug 2023 - Present · thesis in progress

> Thesis: *AI-Driven Object Size Prediction in Visual Data Without Depth Information and Reference Points*

BSc in Computer Engineering (English-medium) Sep 2016 - Aug 2021

PUBLICATIONS

- *Development of Machine Learning-based Revenue Forecasting Models for Electronic Payment and Money Institutions*. IEEE ISMSIT, 2025. DOI.
- *Fault Prediction in Energy Systems: Telemetric Data and Machine Learning Approaches (First Author)*. IEEE ISAS, 2025. DOI.
- *Development of Machine Learning Based Price Quote Prediction Models for Electronic Payment and Money Institutions*. IEEE ISAS, 2025. DOI.
- *Multi-Class Machine Learning Models for Predicting Order Cancellations and Returns in E-Commerce*. IEEE ISAS, 2025. DOI.
- *Forecasting Call Center Arrivals Using Machine Learning*. Osmaniye Korkut Ata University Journal of the Institute of Science and Technology, 2021. DOI.